

OLAYELE GBENGA AWUJOOLA

Data Engineer | Pipeline Architecture | AI-Powered Data Systems

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years building production-grade ETL pipelines, real-time data ingestion systems, and AI-powered automation infrastructure across financial, IoT, and HR analytics domains. Architected a live sensor telemetry pipeline for Lagos State Water Corporation — ingesting MQTT data streams, running time-series anomaly detection, and routing alerts to operational dashboards in production. Designed and automated Python and SQL ETL workflows processing 10,000+ records daily with 35% improvement in data accuracy and 60% reduction in preparation time. Experienced with PostgreSQL, MongoDB, BigQuery, REST API integration, and n8n orchestration, with direct exposure to AI/LLM output pipelines and event-driven data architectures. Operates independently, makes architectural decisions from first principles, and builds scalable data infrastructure that serves both analytical and product-facing use cases.

CORE TECHNICAL SKILLS

Languages & Querying: Python (Pandas, NumPy, Scikit-learn), SQL, R, Google Apps Script

Data Pipeline & ETL: ETL/ELT Pipeline Engineering, Data Modeling, Data Transformation, Data Quality Validation, BigQuery, MQTT

Databases & Warehousing: PostgreSQL, MySQL, MongoDB, SQLite, BigQuery, Airtable

Orchestration & Automation: n8n, Make.com, Zapier, REST API Integration, Webhook & Event-Driven Workflows

AI/ML & LLM Pipelines: Scikit-learn, XGBoost, TensorFlow, PyTorch, OpenCV, LLM Output Processing, NLP, Flask REST API

BI & Visualization: Power BI (DAX, Power Query), Tableau, Apache Superset, Google Sheets, Excel

Infrastructure & DevOps: Git, GitHub, Jupyter Notebook, ThingsBoard IoT Platform, Google Colab

Domain Experience: IoT Sensor Data, Financial Analytics, HR Analytics, AI-Powered Products, Public Sector

PROFESSIONAL EXPERIENCE

Data Engineer / Data Analyst | E-Know Consult | Ibadan, Nigeria

Mar 2024 – Present

- Architected and owned end-to-end ETL pipeline infrastructure processing 10,000+ records daily using Python (Pandas, NumPy) and SQL, enforcing data quality validation gates that reduced downstream error correction by 35% and improved data accuracy by 35% across all downstream BI and analytical systems.
- Designed scalable data architecture and automated reporting pipelines integrating REST APIs, Google Sheets, and internal tools via n8n and Zapier — eliminating 15+ hours of manual processing weekly and reducing task turnaround time across three business units.
- Built and deployed Power BI dashboards with advanced DAX measures and optimized data models, improving reporting efficiency by 40% and enabling real-time decision-making for executive stakeholders across finance, operations, and HR.
- Drove architectural decisions on data tooling selection, pipeline orchestration patterns, and governance documentation — producing runbooks that enabled teams to independently operate, monitor, and maintain all automated data systems with zero engineering dependency.
- Collaborated cross-functionally with Finance, Operations, and HR stakeholders to define project scope, instrument data collection requirements, establish KPI frameworks, and translate pipeline outputs into executive-level business intelligence narratives.

Data Science & Analytics Instructor | SQI College of ICT | Ibadan, Nigeria

Feb 2023 – Feb 2024

- Planned and delivered hands-on data engineering curriculum to 50+ learners — covering Python, SQL, ETL pipeline design, data modeling, Power BI, and workflow automation — achieving a 90%+ project completion rate.

- Supervised medium- to high-complexity data pipeline and analytics projects from scope definition through production delivery, including HR attrition classifiers, financial forecasting pipelines, and e-commerce analytics systems benchmarked on real-world datasets.
- Produced comprehensive governance runbooks, workflow documentation, and configuration guides for all student pipelines, establishing standards for independent operation and repeatability without ongoing instructor dependency.

Research & Data Officer (Part-time) | PIDI-Nigeria | Bauchi, Nigeria

Nov 2020 – Dec 2022

- Collected, transformed, and analyzed structured and unstructured field data from concurrent national programs, ensuring accuracy and consistency for policy reporting and evidence-based decision-making by government stakeholders.
- Mapped end-to-end operational data flows, identified manual bottlenecks, and implemented no-code automation solutions for field teams — reducing manual processing time by 30% and cutting data entry error rates significantly.

KEY ENGINEERING PROJECTS

Smart Water Distribution Analytics System | *Python · MQTT · ThingsBoard · PostgreSQL · Time Series · Geospatial*

- Architected a production real-time data ingestion pipeline for Lagos State Water Corporation — streaming IoT sensor telemetry via MQTT, running time-series anomaly detection to flag pressure drops and pipe fault indicators, and routing structured alerts to operational dashboards consumed by infrastructure engineers in live production.
- Designed scalable ETL workflows for continuous sensor data ingestion, transformation, and warehousing, enabling geospatial device monitoring and live KPI dashboards (flow rate, pressure variance, network uptime) — eliminating manual infrastructure audit cycles for a government utility client.

AI-Powered Applicant Tracking System | *n8n · GPT-4o · Google Drive · PostgreSQL · REST API · Gmail*

- Architected a fully automated 14-node data pipeline on n8n — integrating Google Drive for CV ingestion, building structured data extraction workflows, configuring AI/LLM-driven screening decision logic using GPT-4o, and triggering conditional email notifications — eliminating 8+ hours of manual processing weekly.
- Built structured data logging and analytics outputs to Google Sheets with governance runbooks enabling the hiring team to independently audit, operate, and maintain the pipeline with zero engineering dependency.

HR Attrition Prediction Engine | *Python · Scikit-learn · XGBoost · PostgreSQL · Power BI*

- Designed and built an end-to-end ML data pipeline ingesting 1,470 employee records — engineering features, training a Random Forest + XGBoost ensemble classifier achieving AUC 0.85, and packaging model outputs into segment-level risk score reports for HR stakeholders.
- Built feature importance analysis pipeline surfacing the top 5 attrition drivers (overtime, low satisfaction, early tenure, pay gaps, stagnation), translated into a Power BI dashboard enabling non-technical HR leadership to make evidence-based retention decisions.

Crypto Market Monitoring Automation | *Make.com · CoinGecko API · Airtable · Google Sheets · Telegram*

- Designed and deployed an event-driven market monitoring pipeline integrating CoinGecko API for live price ingestion, configuring conditional alert workflows via Gmail and Telegram, and logging all events to Airtable and Google Sheets with a real-time analytics dashboard.

Social Media Intelligence Monitor | *n8n · SerperDev · X API · Airtable · Slack · Telegram*

- Architected a real-time social media data pipeline on n8n integrating SerperDev and X API for multi-source signal ingestion, defining filtering and data mapping rules, and routing structured intelligence alerts to Telegram and Slack with full Airtable logging for trend analytics — replacing 6+ hours of manual monitoring weekly.

AI-Powered Customer Assistance Chatbot | *Python · NLP · REST API · Webhook Integration*

- Designed and deployed an AI-powered chatbot pipeline integrating NLP capabilities with REST API webhooks — enabling automated query resolution, intelligent response routing, escalation logic, and structured conversation logging to capture interaction data and generate stakeholder performance reports.

E-Commerce Sales Analytics Dashboard | *Power BI · DAX · SQL · Python · Excel*

- Built a multi-page Power BI dashboard with 20+ DAX measures integrating SQL databases and Excel sources, enabling advanced KPI tracking, time-series sales analysis, and RFM-based customer segmentation across product categories and regions for 5,000+ transaction records.

EDUCATION

B.Sc., Computer Science — Federal University, Oye-Ekiti | GPA: 4.2/5.0 2026
HND, Biochemistry — Rufus Giwa Polytechnic, Owo | GPA: 3.3/4.0 2018

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Diploma in Data Science — SQI College of ICT, Ibadan (2023–2024)
- Data Analytics with Power BI — 720 Degree Business School, Ogun (2023)
- Database Management Systems (DBMS) — Aminu Jalam Statistical Software Technology, Bauchi (2022)

RESEARCH & PUBLICATIONS

Published Researcher — UMYU Scientifica (2022) | ORCID: 0000-0002-9432-8984
Ongoing bioinformatics research applying computational data science methods to biological data analysis (Nesprin-1 mutations in cardiomyopathy).